POKHARA UNIVERSITY

Level: Bachelor | Semester : Spring | Year : 2023
Programme: BE | Full Marks: 100
Course: Object Oriented Software Engineering | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What do you understand by the term software engineering? Explain the software process framework in brief. 7
   b) A company is developing a new software product. The product is complex and will require a team of 100 developers to work on it over the course of 2 years. The company is looking for a software engineering model that will allow them to develop the product efficiently and effectively. Propose a software model for this scenario and provide justification for your answer. 8

2. a) What do you understand about outsourcing? Explain suppose an estimated LOC on any software project is 58,600, organizational average productivity is 620 LOC/PM. And the labor rate per month is Rs 12,000. Calculate the total project cost and estimated effort. 7
   b) Explain Risk is an inevitable aspect of any project, particularly software projects. How do project managers identify potential risks, and what are the various methods and tools used for Risk Management, including risk handling strategies, to proactively mitigate and address these risks before they escalate? 8

3. a) Define Requirement Engineering. Explain abstraction, modularity, functional independence and refinement in design concepts. 7
   b) Draw a level-1 DFD diagram for any one of the movie booking systems of Nepal. 8

OR

Draw an activity diagram to borrow books from your respected college library management system.

4. a) How Object-oriented analysis differs from Object-oriented Design? 7
   b) Explain the concept of the Class Diagram along with example of class diagram for Online Shopping System. 8

5. a) Clarify the phrase "Quality has a cost, but lack of quality also has a cost". Explain the defect amplification and removal process with example. 7
   b) What is Black box testing? Demonstrate different types of test you conduct during black box testing. 8

6. a) How can a project manager assure a client that the software product is of high quality? Describe the steps taken to ensure software quality while it is being developed. 7
   b) Describe the SPI framework in detail, outlining its key components and the steps involved in implementing an effective SPI initiative within an organization. 8

7. Write short notes on: **(Any two)** 2×5
   a) User Interface Design
   b) Process Activation Table (PAT)
   c) CMM

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